Basu Tandon

basutandon4257@gmail.com — Ph: 9690612236 — LinkedIn: basu-tandon — Website: basutandon.com

Academic Profile

Year	Degree	Institution	CGPA/%
2021–2023	M.Tech in Materials Science & Engineering	Indian Institute of Technology, Kanpur	8.75/10
2017–2021	B.Tech in Metallurgical & Materials Engineering	National Institute of Technology, Durgapur	8.28/10
2016	Class XII (CBSE)	Gandhi Nagar Public School, Moradabad	75%
2013	Class X (CBSE)	Gandhi Nagar Public School, Moradabad	9.20/10

Professional Experience

Tata Steel Limited, Technologist, Iron Making Technology Group Mar 2024 – Present

- Spearheaded trials and delivered daily, weekly and monthly process reports.
- Conducted extensive analyses of equipment breakdowns and raw material quality fluctuations; provided actionable recommendations for optimizing process parameters to enhance productivity, ensured continuous utilization of process data, and derived novel insights from previously untapped datasets.
- Developed a hybrid numerical model for a coal-based DRI rotary kiln, integrating reaction kinetics and real operational data to predict product quality, coal consumption, and kiln efficiency. Validated simulation results against industrial plant data.
- Implemented a process innovation by decreasing 50^{o}C temperature of DRI reduction zone temperature, securing Rs 30 crore annual savings and longer campaign life. Won the Apex Award at Tata Steel Innovista 2025 innovation awards.
- Sole idea owner and supervisor for implementing INSULAIRR® Li 1000 coating on the DRI kiln shell, reducing heat losses and enhancing thermal efficiency, securing Rs 1.5 Crore annual savings in DRI.
- Authored five technical reports, important ones are mentioned below:
  - Scenario for the Trial of Rice Husk as a Replacement for Coal in DRI Rotary Kiln.
  - Detailed Heat and Mass Balance in DRI. Rotary Kiln for Heat Loss Reduction.
  - Optimizing Sustainability: Utilizing PKSC for Coal-Based DRI Production.

Tata Steel Limited, Management Trainee, Iron Making Technology Group Aug 2023 – Feb 2024

- Worked on process optimization in DRI at Tata Steel, securing 2nd place for outstanding execution of the MT project.

Internships & Conferences

Tata Steel Limited Jan 2024

Participated in the International Conference on Green and Sustainable Iron Making

- Gained insights into sustainable practices in iron making and networked with industry professionals.

Seminar on Sustainability of DRI Industry Sep 26–27, 2024

- Attended a seminar on DRI sustainability, exploring innovations in coal-based DRI, alternative fuels, AI-driven maintenance, and refractory optimization.

IIT Indore May 2019 – Jun 2019

Friction Stir Welding of Aluminum Alloy

- Performed Friction Stir Welding of Al 6061 at IIT Indore, analyzing microstructures to show that forced cooling enhances grain refinement and strength.

Academic Projects

Preparation of Thin films of Al-Doped ZnO and ML-Based Optimization of Processing Parameters Jun 2022 – Jul 2023

Supervisor: Dr. Shikhar Krishn Jha (IIT Kanpur), Dr. Georg Garnweitner (TU Braunschweig)

- Synthesized AZO nanoparticles using solvothermal synthesis and deposited thin films via spin coating.
- Optimized parameters using Genetic Algorithm, achieving minimal resistivity and high transmittance.
- Characterized films using XRD, SEM, TEM, AFM, and UV-Vis Spectroscopy.

Hardness Variation with Carbon Concentration in Steel Jul 2020 – May 2021

Supervisor: Dr. Manas Kumar Mondal (NIT Durgapur)

- Developed a MATLAB model using the Finite Difference Method to simulate surface hardness variations.
- Analyzed the effect of carbon concentration as a function of depth in carburized steel.

Certifications

- NPTEL Nanotechnology Science and Applications (IIT Madras)
- A2 level certification in German Language (2022)
- Structuring Machine Learning Projects - Coursera (DeepLearning.AI)
- Neural Networks and Deep Learning - Coursera (DeepLearning.AI)

Technical Skills

- Programming: Python, MATLAB, C, PowerBI
- Tools: MS Office, Machine Learning, AI Prompt Engineering

Scholastic Achievements

- Awarded DAAD Scholarship, TU Braunschweig, Germany (2022–2023).
- Achieved AIR 420 in GATE Metallurgical Engineering (2021).
- Secured AIR 40799 in JEE Mains (2017) among 1.2 million applicants.